

Notes: Davydiuk, Marchuk & Rosen (JFE, 2024)
Direct lenders in the U.S. middle market

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1 Big picture

- **Question.** What role do direct lenders, especially business development companies (BDCs), play in financing U.S. middle-market firms after bank credit supply contracts?
- **Setting.** BDCs are publicly regulated closed-end investment companies that lend to and invest in small and middle-market private firms while also providing managerial assistance.
- **Data contribution.** The paper hand-collects quarterly BDC investments from SEC filings from 2001Q1 to 2017Q4, covering 69 BDCs, about 10,000 portfolio firms, and over 20,000 debt investments.
- **Identification.** The paper uses three contractionary shocks to traditional lenders: SCAP bank stress tests, the 2009 CIT Group collapse, and FAS 166/167 consolidation rules.
- **Main result.** Counties exposed to these shocks experience higher BDC entry and higher BDC debt investment, implying that BDC capital substitutes for traditional financing.
- **Economic takeaway.** Direct lenders fill a niche for underserved middle-market firms, especially innovative firms, and can stimulate real growth beyond merely replacing bank credit.

2 Data and measurement

2.1 BDC universe and investment-level data

The BDC universe is based on Form N-54A filings with the SEC. Investment data are hand-collected from SEC filings, especially schedules of investments in 10-Q and 10-K filings. The data record portfolio firm names, investment type, deal amount, rate, maturity, industry, and location.

Interpretation. The hand-collected data turn opaque private-credit activity into an observable quarterly panel.

2.2 BDC business model

BDCs must hold at least 70% of investments in eligible assets. They provide debt, equity, warrants, and managerial assistance, while financing themselves with public equity, bonds and notes, revolving credit, and term loans.

Interpretation. BDCs are hybrids: they resemble banks in lending and private equity funds in monitoring and advisory support.

2.3 Shocks and outcomes

The shocks are SCAP, CIT bankruptcy, and FAS 166/167. Outcomes include BDC presence, BDC investment amount, employment growth, patenting, and PD fund activity. Treatment is measured using both indicator and continuous exposure variables.

Interpretation. The shocks identify geographic contractions in traditional credit supply that BDCs can potentially fill.

3 Models and empirical specifications

3.1 County-level difference-in-differences

$$y_{j,t} = \beta Post_t \times Treated_j^{(k)} + \gamma X_{j,t-1} + \eta_j + \tau_t + \varepsilon_{j,t}, \quad k \in \{i, c\}.$$

Interpretation. β measures whether BDC presence or BDC debt investment rises in treated counties after a traditional credit-supply shock.

3.2 Parallel-trends event study

$$y_{j,t} = \sum_t \gamma_t (\lambda_t \times Treated_j) + \gamma X_{j,t-1} + \eta_j + \tau_t + \varepsilon_{j,t}.$$

Interpretation. The event-study coefficients show whether treated counties had similar BDC trends before the shock and how effects evolve afterward.

3.3 Local banking triple difference

$$y_{j,t} = \beta_1 Post_t Treated_j + \beta_2 Post_t LocalBanking_j + \beta_3 Post_t Treated_j LocalBanking_j + \gamma X_{j,t-1} + \eta_j + \tau_t + \varepsilon_{j,t}.$$

Interpretation. A negative β_3 means BDCs enter more strongly where local-bank competition is weak.

3.4 Innovation-county triple difference

$$BDCInvestment_{j,t} = \beta_1 Post_t Treated_j + \beta_2 Post_t e_j + \beta_3 Post_t Treated_j e_j + \gamma X_{j,t-1} + \eta_j + \tau_t + \varepsilon_{j,t}.$$

Interpretation. β_3 tests whether BDC capital flows more strongly to high-tech or high-R&D-intensive counties after credit shocks.

3.5 Firm-level event study

$$y_{f,k,t,t^*} = \beta_h \mathbf{1}\{t \geq t_f^*\} + \eta_f + \tau_t + \theta_{t^*} + \omega_{k,t} + \varepsilon_{f,k,t,t^*}, \quad h \in \{4, 6, 8, 12\}.$$

Interpretation. Firm fixed effects compare firms to themselves before and after BDC investment.

3.6 Staggered dynamic firm design

$$y_{f,k,t,t^*} = \sum_{m=-4}^{+4} \beta_m \mathbf{1}\{t - t_f^* = m\} + \eta_f + \tau_t + \theta_{t^*} + \omega_{k,t} + \varepsilon_{f,k,t,t^*}.$$

Interpretation. This specification checks pre-trends and maps employment and patenting dynamics around the BDC investment date.

3.7 Managerial assistance interaction

$$y_{f,k,t,t^*} = \beta_1 \mathbf{1}\{t \geq t_f^*\} + \beta_2 \mathbf{1}\{t \geq t_f^*\} \times ManagerAttention_{f,t^*} + \eta_f + \tau_t + \theta_{t^*} + \omega_{k,t} + \varepsilon.$$

Interpretation. The interaction tests whether BDC financing has larger real effects when the BDC has more attention to devote to the borrower.

4 Identification and empirical design

The core variation comes from county exposure to three shocks that contracted traditional credit supply but were not caused by local BDC demand. Regressions include county and time fixed effects and lagged local controls. Fig. 7 shows no significant pre-trends for BDC entry or debt investment prior to the shocks. Firm-level designs use firm fixed effects and investment-quarter effects to mitigate selection concerns.

5 Tables and figures: expanded interpretation

This section adds more interpretation to the visual material. Each block emphasizes how to read the result, why the robustness test matters, and how the table or figure connects to the paper’s core mechanism.

5.1 Figures 1–3: growth of BDCs, private debt, and innovation exposure

Read:

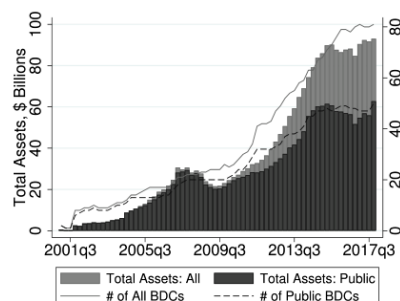
- Figure 1 shows that BDCs grow from a niche financing vehicle into a sizable sector, approaching the scale of a meaningful middle-market credit channel by 2017Q4.
- Figure 2 places BDCs inside the broader private debt market, showing that the paper studies one observable window into a wider direct-lending expansion.
- Figure 3 shows BDC exposure to high-tech and R&D-intensive firms and industries.

Interpretation.

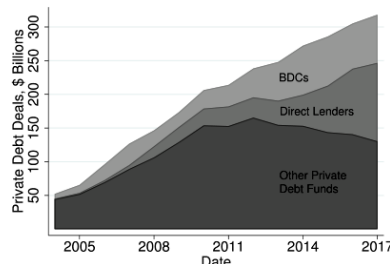
- The three figures establish why BDCs are empirically useful. They are publicly observable, but their borrowers and instruments resemble private middle-market direct lending.
- Figure 3 matters for the later innovation results. If BDCs never invested in innovative sectors, firm-level patenting and high-R&D county tests would be less plausible.
- The figures also motivate the paper’s “migration of credit outside banks” framing: the relevant market is not just BDCs, but the broader private credit ecosystem.

Economic message.

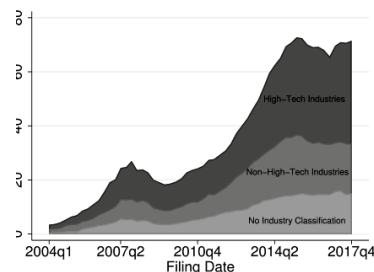
- Direct lending becomes large enough to matter, and BDCs are an observable channel through which private credit reaches opaque middle-market firms.



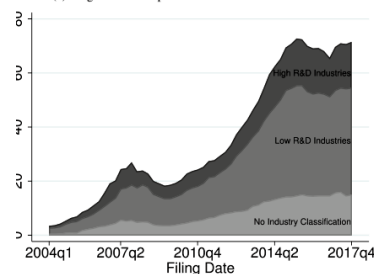
(a) Figure 1: BDC sector growth



(b) Figure 2: broader private debt market



(a): High-Tech Composition of BDC Investments



(b): R&D Composition of BDC Investments

BDC investments in high-tech and high-R&D-intensive industries. The figures industry composition of the BDC investments over time. Panel (a) depicts the of investments allocated to firms in high-tech and non-high-tech industries to the classification of [Hecker \(2005\)](#). Panel (b) depicts the fair value of

(c) Figure 3: high-tech and R&D exposure

5.2 Table 1 and Figures 4–5: BDC portfolio structure and financing model

Read:

- Table 1 shows that BDC portfolios are relatively concentrated and relationship-oriented, rather than anonymous diversified public bond portfolios.
- Figure 4 shows that BDC investments increasingly emphasize debt instruments, especially senior debt, while still retaining flexibility across instruments.
- Figure 5 describes BDC financing: BDCs themselves use debt and equity, but operate under regulatory constraints that keep them more equity-capitalized than banks.

Interpretation.

- This block explains the intermediary technology. BDCs are not merely passive lenders; concentration makes monitoring and advisory support feasible.
- Senior debt exposure connects BDCs to the paper's substitution mechanism: when banks and finance companies pull back, BDCs can step into credit contracts that look loan-like.
- The financing figure matters because BDCs are not traditional deposit-taking banks. Their balance-sheet structure helps explain why they can expand when bank balance sheets are constrained.

Economic message.

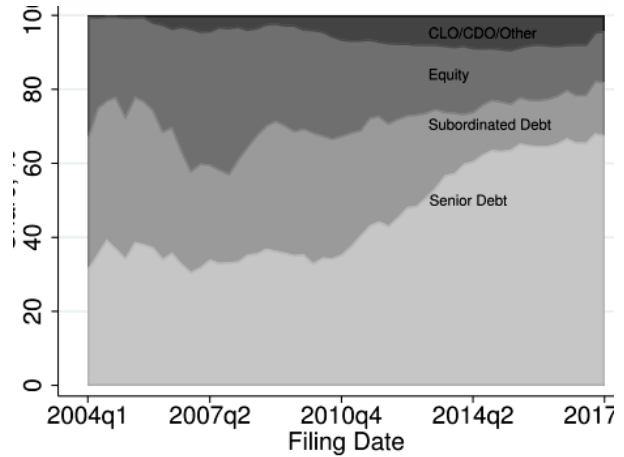
- BDCs combine bank-like lending, private-equity-like monitoring, and nonbank financing. This hybrid structure is central to why they can fill middle-market credit gaps.

Table 1
Investment portfolio of BDCs.

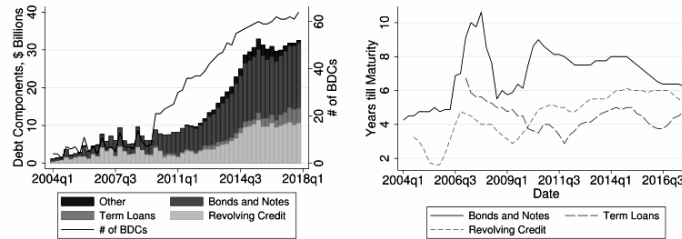
(a) 2010:Q4								
	N	Mean	St.Dev.	Median	10%	25%	75%	90%
Portfolio firms, Count	30	50.93	41.71	42.50	11.50	18.00	67.00	121.50
Allocation top 1 portfolio firms, %	30	16.21	18.33	10.58	4.31	6.22	20.12	30.91
Allocation top 3 portfolio firms, %	30	32.68	22.97	25.21	11.22	16.64	42.53	64.78
Allocation top 5 portfolio firms, %	30	43.69	24.72	34.93	17.22	26.14	57.45	82.32
Allocation top 10 portfolio firms, %	30	61.99	24.89	55.63	28.85	43.08	87.09	97.10
Outstanding deals, Count	30	87.07	76.46	67.50	14.00	36.00	111.00	172.50
Outstanding debt deals, %	30	55.51	26.82	63.08	6.61	47.46	75.00	84.73
Outstanding equity deals, %	30	38.67	25.69	33.45	13.42	18.18	50.00	80.48
Outstanding structured products, %	30	3.84	7.58	0.00	0.00	0.00	3.70	14.28
(b) 2017:Q4								
	N	Mean	St.Dev.	Median	10%	25%	75%	90%
Portfolio firms, Count	59	94.00	183.08	56.00	20.00	31.00	93.00	161.00
Allocation top 1 portfolio firms, %	59	13.04	15.83	8.06	3.98	5.05	14.82	23.40
Allocation top 3 portfolio firms, %	59	25.61	19.23	19.49	8.65	13.28	33.77	45.84
Allocation top 5 portfolio firms, %	59	34.69	21.11	28.41	13.71	19.05	44.60	63.52
Allocation top 10 portfolio firms, %	59	50.86	23.08	46.60	24.25	32.74	63.64	82.60
Outstanding deals, Count	59	151.49	228.82	95.00	34.00	55.00	165.00	214.00
Outstanding debt deals, %	59	64.08	21.17	65.12	32.92	52.78	78.82	88.46
Outstanding equity deals, %	59	32.43	21.95	29.93	6.45	17.39	44.78	67.08
Outstanding structured products, %	59	3.15	8.06	0.00	0.00	0.00	3.45	6.52

The tables report summary statistics for portfolio of BDC investments. Allocations to portfolio firms are recorded in terms of their fair values. The figures represent the cross-sectional statistics across BDCs as of 2010:Q4 and 2017:Q4. Allocations

(a) Table 1: BDC portfolio characteristics

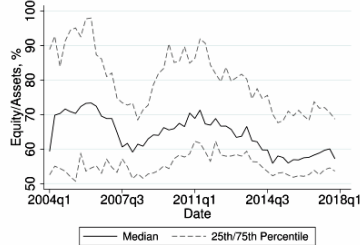


(b) Figure 4: investment instruments



(a): Debt Composition

Panel (b): Debt Maturity



(c): Book Capital Ratio

Figure 5: BDC financing structure

5.3 Figure 6 and Table 2: geography and treated-control balance

Read:

- Figure 6 maps where BDC portfolio firms are located. It shows that BDC activity is not evenly distributed across the United States.
- Table 2 compares treated and control counties before the shocks. It is a diagnostic table for whether exposed counties differ systematically before treatment.

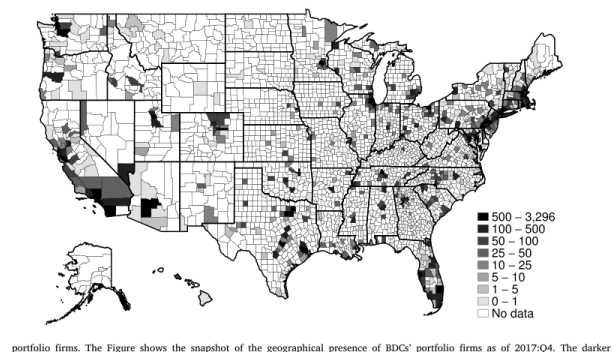
Interpretation.

- The map explains why county fixed effects and local controls are necessary. BDC activity clusters in some regions, so simple cross-sectional comparisons would be misleading.

- Table 2 is not a result table about BDC effects. It is a design table: the reader should use it to assess how much the treated counties differ before shocks and which controls matter.
- Together, the visual and balance table prepare the reader for the difference-in-differences design.

Economic message.

- The empirical design must separate true post-shock BDC entry from pre-existing local credit-market geography.



(a) Figure 6: geography of BDC portfolio firms

Table 2
Descriptive statistics: Treated vs. control counties.

(a): SCAP shock

	Treated			Control			Difference
	N	Mean	St.Dev.	N	Mean	St.Dev.	
Total employment growth, %	1621	0.30	3.23	807	0.13	2.64	0.174
Middle-market employment growth, %	1569	0.43	10.64	681	0.52	6.57	-0.090
Job creation/Total employment, %	1621	4.50	1.46	807	4.76	1.12	-0.262 ***
Job loss/Total employment, %	1621	4.41	1.34	807	4.64	1.02	-0.231 ***
Output growth, %	1600	0.56	5.16	800	1.05	3.74	-0.487 ***
Middle-market firms, %	1637	4.17	1.49	809	3.32	1.45	0.843 ***
# of bank branches per 1000 establishments	1637	16.62	11.61	801	23.91	6.49	-7.284 ***
Deposits HHI	1637	0.25	0.22	801	0.38	0.14	-0.131 ***
Average 2001-2006 HPI growth, %	1588	5.75	2.23	684	4.48	3.20	1.269 ***

(b): CIT shock

	Treated			Control			Difference
	N	Mean	St.Dev.	N	Mean	St.Dev.	
Total employment growth, %	231	-0.44	2.97	1872	-0.53	2.00	0.092
Middle-market employment growth, %	230	-0.81	8.67	1702	-0.25	3.81	-0.558
Job creation/Total employment, %	231	3.99	1.28	1872	4.56	0.58	-0.578 ***
Job loss/Total employment, %	231	4.04	1.18	1872	4.56	0.54	-0.519 ***
Output growth, %	232	-0.78	5.13	1846	-0.35	2.70	-0.430
Middle-market firms, %	236	5.34	1.46	1881	3.74	1.25	1.601 ***
# of bank branches per 1000 establishments	236	13.13	9.68	1873	19.98	3.81	-6.845 ***
Deposits HHI	236	0.19	0.18	1873	0.30	0.11	-0.102 ***
Average 2001-2006 HPI growth, %	236	6.69	2.93	1741	5.21	3.66	1.481 ***

(c): FAS 166/167 shock

	Treated			Control			Difference
	N	Mean	St.Dev.	N	Mean	St.Dev.	
Total employment growth, %	1837	-1.16	3.12	656	-1.09	2.51	-0.072
Middle-market employment growth, %	1767	-1.86	10.97	527	-1.38	6.18	-0.484
Job creation/Total employment, %	1849	4.43	1.54	658	4.68	1.17	-0.246 ***
Job loss/Total employment, %	1849	4.43	1.49	658	4.67	1.10	-0.249 ***
Output growth, %	1807	-0.10	6.04	654	0.91	3.75	-1.008 ***
Middle-market firms, %	1853	4.04	1.46	658	3.13	1.45	0.914 ***
# of bank branches per 1000 establishments	1853	17.58	12.11	650	25.60	7.37	-8.022 ***

(b) Table 2: treated-control county balance

5.4 Table 3 and Figure 7: BDC entry after traditional-credit shocks

Read:

- Table 3 is the first core causal table. It reports positive BDC presence and debt-investment responses after the SCAP, CIT, and FAS 166/167 shocks.
- Figure 7 is the parallel-trends companion. It shows whether treated counties already had rising BDC trends before the shocks.

Interpretation.

- The table supports the substitution story: when traditional bank or finance-company credit supply contracts, BDCs enter or expand in exposed counties.
- The figure is crucial because the main threat is endogenous local demand. If BDCs were already entering treated counties before the shocks, Table 3 would be less persuasive. The absence of strong pre-trends supports the design.
- Using three shocks strengthens the evidence because the result is not tied to a single regulatory or institutional event.

Economic message.

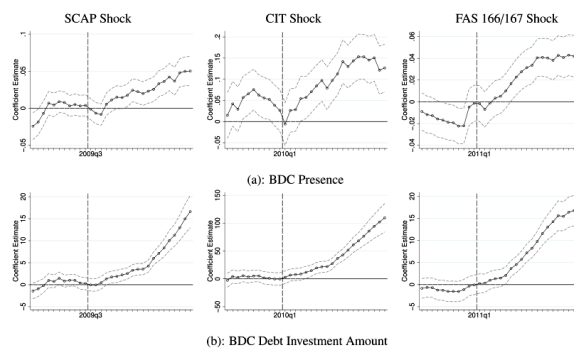
- BDC growth is not just an aggregate post-crisis trend; it is targeted toward places where traditional credit supply falls.

Table 3
BDC entry following credit supply shock.

(a): BDC presence							
	SCAP		CIT		FAS 166/167		Stacked
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Post $\times$ Treated ⁽ⁱ⁾	0.023*** (0.002)		0.055*** (0.007)		0.038*** (0.002)		0.035*** (0.002)
Post $\times$ Treated ⁽ⁱ⁾		0.028*** (0.007)		0.008*** (0.001)		0.085*** (0.006)	
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes	Yes	Yes	No
Time FE	Yes	Yes	Yes	Yes	Yes	Yes	No
County-Cohort FE	No	No	No	No	No	No	Yes
Time-Cohort FE	No	No	No	No	No	No	Yes
R ²	0.72	0.72	0.72	0.72	0.72	0.72	0.72
N	76 078	76 078	65 933	65 933	78 142	78 142	220 153
(b): BDC investment amount							
	SCAP		CIT		FAS 166/167		Stacked
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Post $\times$ Treated ⁽ⁱ⁾	5.255*** (0.252)		41.399*** (1.879)		9.113*** (0.327)		12.841*** (0.363)
Post $\times$ Treated ⁽ⁱ⁾		17.222*** (1.216)		9.964*** (0.501)		46.402*** (2.020)	
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes	Yes	Yes	No
Time FE	Yes	Yes	Yes	Yes	Yes	Yes	No
County-Cohort FE	No	No	No	No	No	No	Yes
Time-Cohort FE	No	No	No	No	No	No	Yes
R ²	0.75	0.75	0.74	0.75	0.73	0.74	0.74
N	76 078	76 078	65 933	65 933	78 142	78 142	220 153

The tables report the estimated coefficients from the difference-in-differences regression:
 $y_{jt} = \beta Post_t \times Treated_j^{(i)} + \gamma X_{j,t-1} + \eta_j + \epsilon_t + \epsilon_{jt}$, $k \in \{1, \epsilon\}$.
The dependent variable is (i) an indicator that equals one if a county j in quarter t has a non-zero debt investment amount by

(a) Table 3: BDC entry and debt investment after shocks



ads for BDC presence and debt investment amount. The figures depict the coefficient estimates of γ s along with the 95% confidence interval $\gamma = \sum_k \gamma_k (X_k \times Treated_j^{(i)}) + \gamma X_{j,t-1} + \eta_j + \epsilon_t + \epsilon_{jt}$, where the dependent variable is (a) an indicator that equals one if a county j in quarter t by BDCs, and (b) the dollar amount of BDC debt investments in county j in quarter t expressed in millions. λ s are post-quarter dummies

(b) Figure 7: event-study / parallel trends

5.5 Tables 4–5: private debt funds and local-bank competition

Read:

- Table 4 shows that private debt funds also enter affected counties. This expands the mechanism beyond the observable BDC sector.
- Table 5 interacts treatment with local banking conditions. BDC investment responds more strongly when local banking is weaker.

Interpretation.

- Table 4 addresses external validity: BDCs are not the entire private debt market, so similar PD fund entry indicates a broader direct-lending response.
- Table 5 identifies where substitution is strongest. If BDCs simply followed all growing counties, local bank weakness would not be especially important. Stronger effects in weak-bank areas support the credit-gap interpretation.
- The local banking interaction also helps distinguish complementarity from substitution. The evidence points to BDCs filling gaps when relationship-bank lending is less available.

Economic message.

- Direct lending acts as a substitute credit channel, especially where local banks cannot or do not absorb the financing gap.

Table 4
PD presence following credit supply shock.

(a): PD presence						
	SCAP		CIT		FAS 166/167	
	(1)	(2)	(3)	(4)	(5)	(6)
Post × Treated ⁽ⁱ⁾	0.074*** (0.002)		0.187*** (0.007)		0.051*** (0.002)	
Post × Treated ^(c)		0.143*** (0.008)		0.034*** (0.001)		0.141*** (0.006)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes
R ²	0.69	0.69	0.73	0.73	0.76	0.76
N	65 692	65 692	47 609	47 609	68 650	68 650
(b): PD investment amount						
	SCAP		CIT		FAS 166/167	
	(1)	(2)	(3)	(4)	(5)	(6)
Post × Treated ⁽ⁱ⁾	278.410*** (13.297)		1317.688*** (70.508)		177.956*** (10.335)	
Post × Treated ^(c)		851.097*** (80.842)		320.048*** (18.722)		1022.846*** (68.381)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes
R ²	0.79	0.79	0.83	0.83	0.82	0.82
N	65 692	65 692	47 609	47 609	68 650	68 650

The tables report the estimated coefficients from the difference-in-differences regression: $y_{jt} = \beta Post_t \times Treated_j^{(i)} + \gamma X_{j,t-1} + \eta_j + \tau_t + \varepsilon_{jt}$. The dependent variable is (i) an indicator that equals one if a county j in quarter t has at least two investment deals by PD funds in Panel (a), and (ii) the dollar amount of PD investments in county j in quarter t expressed in millions in Panel (b). $Post_t$ is a dummy variable that equals one post the capital supply shock starting from (i) 2009:Q3 for the SCAP

(a) Table 4: private debt fund entry

Table 5

BDC investment in counties with local and national banking.

	SCAP		FAS 166/167	
	(1)	(2)	(3)	(4)
Post $\times$ Treated ⁽ⁱ⁾	7.034*** (0.374)	4.916*** (0.240)	11.961*** (0.472)	8.800*** (0.321)
Post $\times$ Local Banking ⁽ⁱ⁾	0.387*** (0.114)		0.691*** (0.137)	
Post $\times$ Treated ⁽ⁱ⁾ $\times$ Local Banking ⁽ⁱ⁾	-4.012*** (0.484)		-6.215*** (0.641)	
Post $\times$ Local Banking ⁽ⁱ⁾		0.327 (0.245)		1.386*** (0.311)
Post $\times$ Treated ⁽ⁱ⁾ $\times$ Local Banking ⁽ⁱ⁾		-8.853*** (0.915)		-14.057*** (1.404)
Controls	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes
R ²	0.75	0.75	0.73	0.73
N	76 060	76 060	78 142	78 142

The tables report the estimated coefficients from the triple difference regression: $y_{j,t} = \beta_1 Post_t \times Treated_j^{(i)} + \beta_2 Post_t \times Local\ Banking_j^{(k)} + \beta_3 Post_t \times Treated_j^{(i)} \times Local\ Banking_j^{(k)} + \gamma X_{j,t-1} + \eta_j + \tau_t + \varepsilon_{j,t}$, for $k \in \{i, c\}$. The dependent variable is the dollar amount of BDC debt investments in county j in quarter t expressed in millions. $Post_t$ is a dummy variable that equals one post the

(b) Table 5: local banking interaction

5.6 Table 6: innovation-oriented county allocation

Read:

- Table 6 asks whether BDC investment flows to counties with high-tech industries or high R&D intensity after traditional credit shocks.
- The key coefficients are treatment interactions with innovation indicators or R&D intensity measures.
- The table shows stronger BDC investment responses in innovation-oriented counties.

Interpretation.

- This is not just a geography result. It links direct lending to the type of firms and local economies that may be hardest for banks to finance after shocks: intangible-intensive, innovative, and growth-oriented borrowers.
- The result also sets up the firm-level patenting analysis. If BDCs expand in innovative counties, it becomes plausible that recipient firms experience innovation-related real effects.
- The table should be read as allocation evidence: BDCs do not only replace missing bank loans mechanically; they direct capital toward counties where innovation financing demand is high.

Economic message.

- Direct lenders appear especially relevant for firms with intangible assets and growth opportunities, where traditional collateral-based lending may be more limited.

Table 6
BDC investment in high-tech and high-R&D-intensive counties.

(a): High-tech counties						
	SCAP		CIT		FAS 166/167	
	(1)	(2)	(3)	(4)	(5)	(6)
Post $\times$ Treated ⁽ⁱ⁾	-0.067 (0.093)	4.301*** (0.218)	-1.892** (0.889)	15.770*** (1.486)	-0.097 (0.095)	7.817*** (0.290)
Post $\times$ High-Tech County ⁽ⁱ⁾	0.514*** (0.129)		1.817*** (0.158)		0.890*** (0.185)	
Post $\times$ Treated ⁽ⁱ⁾ $\times$ High-Tech County ⁽ⁱ⁾	8.744*** (0.449)		48.383*** (2.332)		15.358*** (0.602)	
Post $\times$ High-Tech County ⁽ⁱ⁾		0.025*** (0.007)		0.111*** (0.015)		0.016* (0.008)
Post $\times$ Treated ⁽ⁱ⁾ $\times$ High-Tech County ⁽ⁱ⁾		1.135*** (0.071)		5.722*** (0.352)		1.866*** (0.096)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes
R ²	0.74	0.74	0.73	0.74	0.74	0.74
N	73 904	73 904	63 934	63 934	76 062	76 062

(b): High-R&D-intensive counties						
	SCAP		CIT		FAS 166/167	
	(1)	(2)	(3)	(4)	(5)	(6)
Post $\times$ Treated ⁽ⁱ⁾	0.102 (0.083)	4.910*** (0.247)	-2.953*** (0.724)	35.426*** (1.966)	-0.367*** (0.091)	8.474*** (0.319)
Post $\times$ R&D County ⁽ⁱ⁾	0.801*** (0.137)		1.774*** (0.159)		0.501*** (0.192)	
Post $\times$ Treated ⁽ⁱ⁾ $\times$ R&D County ⁽ⁱ⁾	8.164*** (0.443)		49.868*** (2.284)		15.526*** (0.594)	
Post $\times$ R&D County ⁽ⁱ⁾		0.008 (0.007)		0.097*** (0.014)		-0.004 (0.008)
Post $\times$ Treated ⁽ⁱ⁾ $\times$ R&D County ⁽ⁱ⁾		0.676*** (0.049)		2.855*** (0.384)		1.220*** (0.076)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes
R ²	0.74	0.74	0.73	0.73	0.74	0.74
N	73 904	73 904	63 934	63 934	76 062	76 062

The tables report the estimated coefficients from the triple difference regression:
 $y_{ijt} = \beta_1 Post_{it} \times Treated_{it}^{(i)} + \beta_2 Post_{it} \times e_{it}^{(i)} + \beta_3 Post_{it} \times Treated_{it}^{(i)} \times e_{it}^{(i)} + \gamma X_{ijt,t-1} + \eta_j + \tau_t + \epsilon_{ijt}$, for $k \in \{i, c\}$.
The dependent variable in each regression is the dollar amount of BDC debt investments in county j in quarter t expressed in millions. $Post_{it}$ is a dummy variable that equals one post the capital supply shock starting from (i) 2009:Q3 for the SCAP shock, and (ii) 2011:Q1 for the FAS 166/167 shock. $e_{it}^{(i)}$ is a dummy variable that equals one if a county j has the above median employment share in high-tech and high-R&D-intensive industries in Panels (a) and (b), respectively. $e_{it}^{(i)}$ is the employment share in high-tech and high-R&D-intensive industries in Panels (a) and (b), respectively. The set of treated counties is defined in Section 3.1. The control group includes non-treated counties with BDC debt financing during the pre-shock period along with non-treated counties adjacent to them. The sample covers the period three years before and five years after the shock.
***, **, and * indicate significance at the 1%, 5%, and 10% level.

Table 6: BDC investment in high-tech and high-R&D counties

5.7 Tables 7–8 and Figure 8: firm-level real outcomes

Read:

- Table 7 reports firm-level outcomes after BDC financing, including employment growth and patenting.
- Table 8 uses a staggered design to map outcomes around BDC investment dates.
- Figure 8 visually checks pre-trends and the post-treatment trajectory for firm outcomes.

Interpretation.

- The firm-level results move the paper from “BDCs enter shocked counties” to “BDC financing is associated with real firm growth.”
- Employment effects are relatively direct: BDC capital can relax financing constraints and support expansion. Patenting effects are more delayed and connect to innovation financing.
- The pre-trend evidence is essential because BDCs may select firms already on a growth path. Firm fixed effects and event-study plots reduce, though do not eliminate, this selection concern.

Economic message.

- BDC financing is associated with real activity, not merely a reshuffling of financial claims.

Table 7

Firm outcomes and BDC financing: Event-study analysis.

(a): Employment growth					
	h = 4	h = 6	h = 8	h = 12	Placebo
$1\{t \geq t_f^*\}$	0.901*** (0.281)	1.084*** (0.360)	1.197*** (0.396)	0.800* (0.418)	
$1\{t \geq t_f^* - 8\}$					0.469 (0.364)
Investment Qtr FE	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes
Industry-Year FE	Yes	Yes	Yes	Yes	Yes
R2	0.626	0.587	0.555	0.506	0.545
N	7100	8674	10 256	13 412	9633

(b): Patenting propensity					
	h = 4	h = 6	h = 8	h = 12	Placebo
$1\{t \geq t_f^*\}$	-0.000 (0.013)	0.010 (0.012)	0.020* (0.011)	0.021* (0.011)	
$1\{t \geq t_f^* - 8\}$					-0.001 (0.011)
Investment Qtr FE	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes
Industry-Year FE	Yes	Yes	Yes	Yes	Yes
R2	0.737	0.731	0.726	0.700	0.698
N	7681	9385	11 097	14 513	11 097

The tables report the estimated coefficients from the firm-level regression:

$$y_{f,k,t,t^*} = \beta_0 1\{t \geq t_f^*\} + \beta_1 1\{t \geq t_f^* - 8\} + \eta_f + \tau_t + \theta_{k,t} + \omega_{k,t} + \varepsilon_{f,k,t,t^*} \quad \text{for } h \in \{4, 6, 8, 12\}$$

(a) Table 7: firm-level event-study outcomes

Table 8

Firm outcomes and BDC financing: Staggered difference-in-differences analysis.

	Employment growth		Patenting	
	Main	Placebo	Main	Placebo
$1\{t - t_f^* = -4\}$	-0.191 (0.233)	-0.207 (0.309)	0.004 (0.015)	0.003 (0.012)
$1\{t - t_f^* = -3\}$	-0.023 (0.413)	-0.608 (0.523)	0.009 (0.018)	-0.014 (0.017)
$1\{t - t_f^* = -2\}$	-0.106 (0.607)	-0.876 (0.728)	0.002 (0.022)	-0.016 (0.021)
$1\{t - t_f^* = -1\}$	-0.003 (0.801)	-1.057 (0.935)	0.004 (0.026)	-0.002 (0.026)
$1\{t - t_f^* = 0\}$	0.683** (0.329)	-0.066 (0.298)	0.006 (0.013)	-0.012 (0.012)
$1\{t - t_f^* = +1\}$	0.841* (0.460)	-0.193 (0.435)	0.021 (0.015)	0.001 (0.014)
$1\{t - t_f^* = +2\}$	0.532 (0.581)	-0.432 (0.564)	0.021 (0.018)	-0.017 (0.016)
$1\{t - t_f^* = +3\}$	0.350 (0.713)	-0.813 (0.696)	0.049** (0.019)	-0.023 (0.019)
$1\{t - t_f^* = +4\}$	0.266 (0.849)	-1.050 (0.816)	0.033 (0.022)	-0.018 (0.021)
Investment Qtr FE	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes
Industry-year FE	Yes	Yes	Yes	Yes
N	8357	6436	10 240	10 240

The tables report the estimated coefficients from the difference-in-differences regression with staggered treatment adoption:

$$y_{f,k,t,t^*} = \sum_{m=-4}^{m=+4} \beta_m 1\{t - t_f^* = m\} + \eta_f + \tau_t + \theta_{k,t} + \omega_{k,t} + \varepsilon_{f,k,t,t^*}$$

The dependent variable is (a) the employment growth and (b) the natural logarithm of one plus the number of patents filed by a firm f in an industry k in a quarter t with an investment quarter t^* . t_f^* is the BDC investment quarter in firm f . m is the number

(b) Table 8: staggered DID outcomes

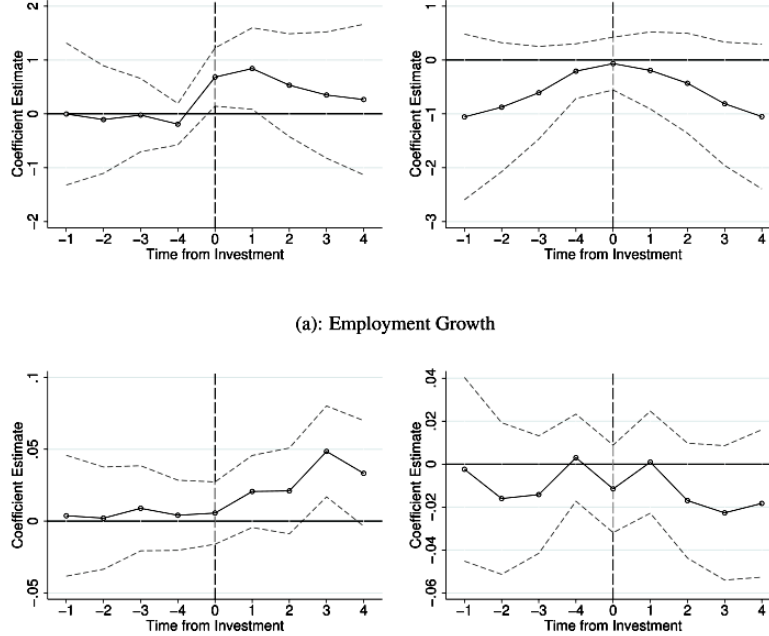


Figure 8: firm-level parallel trends

5.8 Tables 9–11: managerial attention, PD comparison, and shock-window robustness

Read:

- Table 9 interacts BDC financing with managerial attention. Effects are larger when the BDC has more attention to devote to the borrower.
- Table 10 compares BDC-financed firms to firms financed by private debt funds.
- Table 11 narrows identification around shock windows and tests whether the real effects line up with the credit-supply shocks.

Interpretation.

- Table 9 is the clearest evidence that BDCs may provide more than capital. If managerial attention amplifies employment outcomes, monitoring/advice is part of the production technology.
- Table 10 helps separate what is BDC-specific from what is general private-credit financing. If PD-financed firms show similar effects, the mechanism is broader private debt; if BDCs differ, the BDC structure matters.
- Table 11 tightens the causal story by connecting firm outcomes more closely to credit-supply shock exposure rather than generic post-investment growth.

Economic message.

- The real effects of direct lending likely come from a package: capital, monitoring, advice, and the ability to lend when traditional intermediaries retreat.

Table 9

Firm employment growth and BDC managerial assistance.

	h = 4	h = 6	h = 8	h = 12	Placebo
$\mathbb{1}\{t \geq t_f^i\}$	0.909***	1.102***	1.220***	0.825**	
	(0.281)	(0.360)	(0.396)	(0.419)	
$\mathbb{1}\{t \geq t_f^i\} \times \text{Manager Attention}$	0.331	0.426*	0.436*	0.404*	
	(0.229)	(0.221)	(0.234)	(0.231)	
$\mathbb{1}\{t \geq t_f^i - 8\}$				0.464	
				(0.364)	
$\mathbb{1}\{t \geq t_f^i - 8\} \times \text{Manager Attention}$				-0.129	
				(0.243)	
Investment Qtr FE	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes
Industry-Year FE	Yes	Yes	Yes	Yes	Yes
R2	0.626	0.588	0.556	0.506	0.545
N	7100	8674	10256	13412	9633

The tables report the estimated coefficients from the firm-level regression:

$$y_{f,k,t} = \beta_{0,k} \mathbb{1}\{t \geq t_f^i\} + \beta_{1,k} \mathbb{1}\{t \geq t_f^i\} \text{Manager Attention}_{f,t} + \eta_f + \tau_t + \theta_{i,t} + \omega_{k,t} + \epsilon_{f,k,t},$$

for $h \in \{4, 6, 8, 12\}$.

The dependent variable is the employment growth of a firm f in an industry k in a quarter t with an investment quarter t_f^i . $\mathbb{1}\{t \geq t_f^i\}$ is an indicator variable which equals one for quarters following the BDC investment quarter in a firm f , and zero otherwise. *Manager Attention* _{f,t} is the inverse of the number of firms in the portfolio of the BDC when it initiates an investment in a firm f . We standardize the *Manager Attention* variable by subtracting its mean and scaling by its standard deviation. In columns (1)–(4), the sample period includes four quarters prior to the investment date and $h \in \{4, 6, 8, 12\}$ quarters after the investment date for each firm f . The specification *Placebo* in the rightmost column is similar to the specification $h = 8$ with the sample period and investment date shifted backwards by eight quarters. ***, **, and * indicate significance at the 1%, 5%, and 10% level.

Table 10

Firm employment growth and PD financing: Event-study analysis.

	h = 4	h = 6	h = 8	h = 12	Placebo
$\mathbb{1}\{t \geq t_f^i\}$	0.724	1.189**	1.408***	1.228**	
	(0.452)	(0.503)	(0.525)	(0.521)	
$\mathbb{1}\{t \geq t_f^i\} \times \text{BDC Funded}$	0.195	-0.029	-0.160	-0.496	
	(0.617)	(0.648)	(0.664)	(0.656)	
$\mathbb{1}\{t \geq t_f^i - 8\}$				0.447	
				(0.503)	
$\mathbb{1}\{t \geq t_f^i - 8\} \times \text{BDC Funded}$				0.267	
				(0.648)	
Investment Qtr FE	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes
Industry-Year FE	Yes	Yes	Yes	Yes	Yes
R2	0.651	0.602	0.567	0.511	0.581
N	12115	14737	17334	22358	16914

The tables report the estimated coefficients from the firm-level regression:

$$y_{f,k,t} = \beta_{0,k} \mathbb{1}\{t \geq t_f^i\} + \beta_{1,k} \mathbb{1}\{t \geq t_f^i\} \text{BDC Funded}_{f,t} + \eta_f + \tau_t + \theta_{i,t} + \omega_{k,t} + \epsilon_{f,k,t},$$

for $h \in \{4, 6, 8, 12\}$. The dependent variable is the employment growth of a firm f in an industry k in a quarter t with an investment quarter t_f^i . $\mathbb{1}\{t \geq t_f^i\}$ is an indicator variable which equals one for quarters following the BDC or PD investment quarter in a firm f , and zero otherwise. *BDC Funded* _{f,t} is an indicator variable which equals one if a firm f receives financing from BDCs and zero if a firm f receives financing from PD capital providers. In columns (1)–(4), the sample period includes four quarters prior to the investment date and $h \in \{4, 6, 8, 12\}$ quarters after the investment date for each firm f . The specification *Placebo* in the rightmost column is similar to the specification $h = 8$ with the sample period and investment date shifted backwards by eight quarters. ***, **, and * indicate significance at the 1%, 5%, and 10% level.

	SCAP	CTT	FAS 166/167	Stacked
Post	1.084*	0.969*	0.091	0.876***
	(0.632)	(0.515)	(0.406)	(0.246)
Post $\times$ Treated ⁽ⁱ⁾	-1.582**	-0.400	-1.525***	-1.108***
	(0.674)	(0.463)	(0.413)	(0.281)
Controls	Yes	Yes	Yes	Yes
Investment Qtr FE	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	No
Firm-Cohort FE	No	No	No	Yes
R ²	0.60	0.52	0.51	0.53
N	2496	4117	4605	11218

The tables report the estimated coefficients from the difference-in-differences regression:

$$\Delta \text{emp}_{f,k,t} = \beta_1 \text{Post}_t + \beta_2 \text{Post}_t \times \text{Treated}_t^{(i)} + \gamma X_{f,t-1} + \eta_f + \theta_{i,t} + \epsilon_{f,k,t},$$

The dependent variable is the employment growth of a firm f located in a county j in an industry k in quarter t with an investment quarter t_f^i . *Post* _{t} is a dummy variable that equals one post the capital supply shock starting from (i) 2009:Q3 for the SCAP shock, (ii) 2010:Q1 for the CTT shock, and (iii) 2011:Q1 for the FAS 166/167 shock. *Treated* _{f} ⁽ⁱ⁾ is equal to one if a county j 's treatment intensity is above the 25th percentile of *Treated*⁽ⁱ⁾ and zero otherwise. The county treatment intensity is defined in Section 3.1.

(a) Table 9: managerial attention

(b) Table 10: private-debt comparison

(c) Table 11: shock-window real effects

6 Short summary

- The paper studies direct lending in the U.S. middle market using BDCs as an observable window into private debt.
- BDCs expanded rapidly after the early 2000s and reached close to \$100 billion in assets by 2017Q4.

- Counties exposed to SCAP, CIT, and FAS 166/167 shocks see more BDC presence and debt capital.
- Private debt funds show similar entry patterns, suggesting broad private-debt substitution for bank credit.
- BDC capital flows especially to high-tech and high-R&D-intensive counties.
- Firms receiving BDC financing exhibit higher employment growth and patenting activity.
- Managerial assistance appears to amplify the employment effects of BDC financing.

7 Conclusion

7.1 Core conclusion

Davydiuk, Marchuk and Rosen show that direct lenders, especially BDCs, act as substitutes for traditional financial institutions when bank and finance-company credit supply contracts. Their capital flows into affected counties, especially innovative areas, and is associated with employment growth and patenting at recipient firms.

7.2 Economic intuition

The economic intuition is that middle-market firms are informationally opaque and too small for public capital markets but often too risky or tailored for banks after regulatory tightening. BDCs combine credit provision with monitoring, flexibility, and managerial assistance. This makes them well suited to fill credit gaps left by constrained traditional lenders.

7.3 Expanded conclusion: what the paper changes conceptually

The paper’s conclusion is not only that BDCs grew. It is that direct lenders can substitute for constrained banks and finance companies in the middle-market credit channel, and that this substitution can have measurable real effects.

- The paper makes private credit empirically visible by using BDCs, which disclose detailed portfolios while operating in the same broad direct-lending market as private debt funds.
- The county-level shock design shows that nonbank credit expands where traditional lenders become constrained, rather than merely expanding everywhere during a private-credit boom.
- The firm-level outcomes connect credit supply to real activity: BDC financing is associated with employment growth and patenting, not only with loan origination.
- The managerial-attention tests suggest that direct lenders may supply monitoring and advice in addition to capital.

7.4 Mechanism: why direct lenders can fill the gap

Middle-market firms are often too opaque for public debt and too small or risky for banks after regulatory or balance-sheet shocks. Direct lenders are designed to operate in exactly this space.

- BDCs hold concentrated portfolios, which makes monitoring and borrower relationships feasible.
- Their investment mix is loan-like and often senior-secured, so they can replace part of the bank-credit function.
- Their nonbank funding structure can allow expansion when banks are constrained by regulation, capital pressure, or balance-sheet repair.

- The innovative-county and patenting results indicate that direct lenders can finance intangible-intensive firms that traditional collateral-based lending may underserve.

7.5 How the evidence chain supports the conclusion

The paper’s conclusion is cumulative: each empirical block supports a different link in the direct-lending mechanism.

- Figures on sector growth establish that BDCs and private debt became large enough to matter for middle-market credit.
- Portfolio tables show that BDCs are debt-oriented and relationship-oriented, not merely diversified public securities vehicles.
- SCAP, CIT, and FAS 166/167 shock tests identify entry where traditional lenders faced supply contractions.
- Private debt comparisons show that the response is not unique to the BDC legal form; BDCs are an observable proxy for a broader private-credit substitution channel.
- Firm event studies and DiD tests move from capital flows to real effects, while managerial-attention results help explain why effects may be larger for some borrower-lender relationships.

7.6 Policy and corporate-finance implications

The findings complicate a simple bank-versus-nonbank view. Direct lenders can soften the real effects of bank credit contractions, but they also move credit intermediation into a less traditional regulatory perimeter.

- For corporate finance, the paper highlights a source of financing for firms that are too opaque for public debt but may still have high-growth or innovation projects.
- For banking policy, the paper shows that restricting bank balance sheets can shift credit activity toward nonbanks rather than only reducing credit quantities.
- For innovation finance, the results suggest that private credit can support employment and patenting when direct lenders combine capital with monitoring and assistance.
- For financial stability, the paper raises a broader question: credit gaps may be filled, but risk may migrate to intermediaries with different disclosure, funding, and regulatory structures.

7.7 Final takeaway

Direct lending is a substitute credit technology with real consequences. When traditional lenders pull back, BDCs and private debt funds can redirect capital toward affected middle-market firms, especially innovative borrowers. The central lesson is that modern credit intermediation is increasingly multi-channel: bank constraints do not only reduce credit; they also reshape where credit is produced and monitored.